Trend Ratio-based Portfolio Optimization Model Adopting Entanglement-enhanced Quantum-inspired Evolutionary Computation in the Global Financial Markets

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***Abstract*—Financial management is a critical and complicated issue. People tend to invest in the stock market and expect a stable uptrend portfolio to spread the investment risk effectively. As global economies affect each other, investing in global markets is the best way to address systematic risk for effective financial management. Selecting an appropriate portfolio across global markets involves a substantial solution space; thus, an efficient and effective evolutionary computation is proposed. The novel entanglement-enhanced quantum-inspired optimization technique is proposed as an efficient mechanism to search for the relationship between stocks with high-dependency solutions. The novel indicator trend ratio aims to evaluate the investor-desired portfolio with a stable uptrend under a perfect balance between return and risk. Therefore, this study is the first attempt to evaluate the major global markets in the Group of Seven (G7) countries to expand the generality of trend ratio usage. The experiments demonstrate that the proposed novel approach has better and more precise searchability than other state-of-the-art optimization. The proposed intelligent model outperforms the Sharpe ratio, benchmark strategy, and index performance regarding the trend ratio, risk, maximum drawdown, and profit factor. The results reveal that risk dispersion substantially improves investment performance. Under the evaluation of the trend ratio, the proposed system integrates G7 markets as an expansion of a prospective avenue for global asset management.**

***Index Terms*—Trend ratio, Portfolio selection, Global market, Quantum-inspired optimization, Local search.**

I. Introduction

Economic globalization is a feature of modern society. Global economies are mutually influenced by each other. The effects of the global COVID-19 pandemic provide clear evidence. The economy has experienced a severe plunge worldwide. Financial management enables people to cope with this kind of unforeseen emergency. The stock market is a major financial management option, providing advantages such as a low investment threshold, high market liquidity, and public and transparent information. Furthermore, the stock market has a leading role in reflecting the economics of a country.

Investments should consider profit and risk simultaneously. High profit is accompanied by high risk, and low risk is accompanied by low profit. Modern portfolio theory (MPT) [1] indicates that a portfolio can greatly diversify financial risk. The construction of a balanced portfolio to diversify assets is a complex problem. The trade-off indicator is used to evaluate the relationship between return and risk. The Sharpe ratio [2] is a commonly used indicator to assess portfolio return and risk. The Sharpe ratio is the expected return per unit of risk. The Sharpe ratio inherits the core concept of MPT and assumes that investors are risk-averse; i.e., when two portfolios have the same return, investors would choose the one with lower risk. In the Sharpe ratio, the return uses the risk premium, and the variance represents the risk. According to this risk definition, a steady downtrend portfolio is at high risk because it continuously falls and deviates from the average, but a stable uptrend portfolio would be identified as having the same high risk. To overcome this issue, the trend ratio has been proposed to evaluate the performance of portfolios following investor psychology.

The trend ratio calculates the portfolio's daily expected return by its daily risk. The primary distinction is that the trend ratio assesses portfolio risk based on the trend line acquired by linear regression. Deviation of the portfolio trend from the trend line is regarded as portfolio risk, and the trend line's slope represents the portfolio return. When two portfolios have identical daily expected returns, the lower-risk portfolio will have a higher trend ratio. Utilizing the trend ratio, investors can identify a stable uptrend portfolio as a low-risk, high-return portfolio that balances return and risk.

The trend ratio is a novel indicator to evaluate portfolio performance in a manner expected by investors [3, 4]. This study aims to expand the usage and prove the trend ratio's generality across stock markets. Seven major industrialized countries have formed an intergovernmental forum to discuss their economy and currency policy (G7). G7 members lead global affairs and influence half of the world's economy, accounting for 44% of the global gross domestic product (GDP) according to a World Bank report in 2021. The markets of the G7 are very representative to demonstrate the common usage and applicability of the trend ratio.

Different markets have unique industry compositions. Popular and growing online trading platforms have made globalized financial management easier. In addition to the separate G7 markets, investment across markets is a future trend in stabilizing investment efficiency. Thus, this study discusses the separate performance of G7 markets and integrates G7 markets into a global market to reduce systematic risk to an extreme extent. As the solution space is enormous, effective computational intelligence is required to construct a high-efficiency portfolio. Evolutionary computation (EC) is a popular method for quickly searching for the optimal portfolio.

The quantum-inspired tabu search algorithm (QTS) is a powerful EC approach that excels at combinatorial optimization [5], numerical function optimization [6], and financial issues [7]. QTS can be near the best solution and far from the worst solution simultaneously by simulating quantum superposition. In a global market, the promotion of QTS's search efficiency is required to produce a near-optimal solution. To improve QTS's exploitation and exploration abilities, global best solution guided and quantum-not gate mechanisms are proposed, resulting in GNQTS [4, 8]. In this study, entanglement-enhanced local search mechanisms based on

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GNQTS are proposed in a method named EL-GNQTS. Local search is an efficient method for elevating the solution quality around a local optimum. The entanglement mechanism can be used to model the information-based local search mechanism according to portfolio optimization characteristics, which have not been explored previously. EL-GNQTS is a novel approach to modeling the entanglement relation as the neighborhood to enhance solution quality with low computation overhead.

Training data and its length are also important for financial issues. A training period that is too long or too short is inappropriate. The sliding window mechanism updates data over time to retain the freshness of the data. The frequency of rebalancing depends on the sliding window period: monthly sliding windows rebalance more often than annual ones. Aside from the continuous sliding window, the year-on-year sliding window is taken into account when the industry has a business cycle. The year-on-year sliding window reveals industry correlations in the investment target.

In summary, this study proposes a novel concept for portfolio optimization: an entanglement-enhanced local search optimization technique to effectively search for potential neighborhood areas to improve solution quality. Besides, this study is the first attempt to expand the usage of the trend ratio model in G7 multiple markets at a time and discuss their similarities and discrepancies. This study provides a general and comprehensive discussion of these markets to enhance the utility of the trend ratio model. Moreover, investment across markets provides a more robust investment model for future portfolio optimization.

II. Related Work

Investment is a complicated problem in the real world. People aim to maximize investment returns and minimize investment risk. There are three major branches: portfolio optimization, market timing [9], and stock price forecasting [10]. This paper focuses on the portfolio optimization problem to identify portfolios with high returns and low risk.

Modern portfolio theory (MPT) is a well-known investment theory that assumes investors are risk-averse and emphasizes the importance of simultaneously considering investment risk and profit. Investing in multiple targets and constructing a portfolio can efficiently mitigate risk. Investors commonly use assessment indicators to evaluate their portfolio performance. One famous assessment indicator is the Sharpe ratio, which defines risk based on MPT. Risk in MPT is represented as the variance, which is the deviation from the mean value. Typically, the Sharpe ratio has been employed to analyze portfolio performance [11]. One study [12] uses a fuzzy system to investigate portfolio optimization issues and further proposes the fuzzy Sharpe ratio, which can service fuzzy systems. Due to the Sharpe ratio's risk definition, the upside portfolio trend cannot be exactly evaluated. The Sortino ratio [13] is an indicator that simultaneously considers investment return and risk according to the definition of the Sharpe ratio, but only falling fluctuation is included in the risk. Study [14] uses deep reinforcement learning with the Sortino ratio to search for a risk-aware portfolio with a higher return.

There is an innovative indicator called the trend ratio [3] to address the risk definition issue and assess upside portfolio performance more accurately. The trend ratio is developed based on the funds standardization (FS) concept, proposed by study [11] to convert stock prices into daily fund variations that precisely reflect investors' emotions. The trend ratio aims to find a stable uptrend portfolio through the trend line, where the slope of the trend line represents the daily expected return and deviation from the trend line is defined as the risk. Studies [3, 4, 8, 15] assess market performance via the trend ratio. The excellent results demonstrate the utility of the trend ratio in constructing an upward portfolio with a steady trend, which makes it worthy of extending its usage into representative markets. Therefore, the trend ratio is adopted in this paper to identify portfolios with a stable uptrend in the G7 markets.

Study [16] shows that artificial intelligence applications are becoming increasingly common in financial investment, including portfolio optimization, to eliminate momentary irrationality. The solution space of portfolio optimization is substantial due to the large number of stocks in stock markets. As portfolio construction searches within a very large space, many EC techniques are widely adopted, such as the genetic algorithm (GA) [11], particle swarm optimization (PSO) [17], artificial bee colony algorithm (ABC) [18], level-based learning swarm optimizer [19], multi-objective evolutionary algorithm (MOEA/D) [20], and quantum-inspired tabu search algorithm (QTS) [6].

QTS-related techniques demonstrate their exploration and exploitation search ability in various problems, such as in the 0/1 knapsack problem [5], reversible circuits [21], and wormhole attack detection [22]. GNQTS improves QTS and has a strong ability to search for potential solutions in a huge solution space. Some studies [4, 8, 15] show that the GNQTS has superior performance in portfolio selection problems. According to these studies, EC presents an excellent capability to identify potential portfolio solutions. For better convergence in the considerable solution space, a local search mechanism is commonly embedded in EC to promote search efficiency [23]. This study integrates the GNQTS and the entanglement local search concept to propose an effective algorithm, EL-GNQTS, to search the potential area precisely.

The global economy is tied to each nation's economy. In recent years, a growing number of investors have invested in the stock markets of other countries to spread investment risk. The G7 group comprises the seven largest industrialized nations and the world's major economic systems. The G7 deserves special scrutiny because it is the industrial leader of the industrialized world. Study [9] discusses the performance of 10 international financial markets, including some G7 countries. Study [24] analyzes how financial contagion happens between China and G7 countries during COVID-19 in financial and nonfinancial enterprises. Thus, G7 markets are important targets to prove the universality of an innovative indicator in finance.

To the best of our knowledge, entanglement local search for portfolio optimization has not been explored previously. The proposed EL-GNQTS is a novel contribution to define the entanglement relation in portfolio optimization and accurately searches for potential areas rather than randomly searching. Additionally, this is the first attempt to demonstrate the generalizability of the trend ratio by discussing the relations of multiple financial markets. Furthermore, we are the first to further integrate multiple financial markets into a global

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market via the trend ratio model assessment.

III. Proposed Method

This study proposes an intelligent system to construct a stable uptrend portfolio efficiently in a suitable period. The system can be applied in each major market of the G7 to exhibit its effectiveness. Three main components are introduced in this section: a novel trend ratio indicator to evaluate the stable uptrend portfolio in Section III-A, the proposed entanglement-enhanced local search mechanism based on GN-QTS optimization (EL-GNQTS) to search for the near-optimal high trend ratio portfolio in Section III-B, and the sliding window mechanism to retain fresh data and find a suitable investment time in Section III-C.

A. Trend Ratio

The trend ratio (TR) is a novel assessment metric accommodating investors' expectations in a stable uptrend portfolio. The TR measures the daily expected return divided by daily risk, indicating the return per unit of risk, as shown in Equation (1). The novel TR uses the trend line to evaluate portfolio risk: a stable uptrend portfolio has less deviation from the trend line and thus has lower risk. The TR is based on FS [11] and transforms the daily stock price into daily fund fluctuation. Regression of the daily FS yields the portfolio trend line (TL) with a fixed y-intercept, as shown in Equation (2). The slope of the trend line is the daily expected return. The volume of daily FS deviation from the trend line is the risk, as shown in Equations (3) and (4), where FS_d is the portfolio FS on the d_{th} day, N denotes the initial funds, and td is the total number of days.

$$TR = \frac{Daily\ Expected\ Return\ (m)}{Daily\ Risk} \tag{1}$$

$$TL_d = (m \cdot\ d)\ + N \tag{2}$$

$$m = \frac{\sum[(FS_d\ \cdot\ d) - (N \cdot d)]}{\sum(d)^2} \tag{3}$$

$$Daily\ Risk = \sqrt{\frac{\sum(FS_d - TL_d)^2}{td}} \tag{4}$$

B. Entanglement-enhanced Local Search based on Global-best Guided Quantum-inspired Tabu Search Algorithm with Quantum-Not Gate (EL-GNQTS)

The G7 stock market has thousands of stocks. Finding an optimal solution in such a huge solution space is difficult. Hence, this study proposes EL-GNQTS, which inherits the characteristics of QTS and has global best solution guidance and a quantum-not gate to promote the exploitation and exploration abilities of the algorithm. The entanglement-enhanced local search further improves the solution quality via an economic search around local optima through the entanglement-like relation. Fig. 1 shows a flow chart of the proposed EL-GNQTS. The following discusses the detailed steps of implementation.

1) Problem Encoding: EL-GNQTS uses the binary 0 or 1 to represent the selection state of each stock in the portfolio, S_{si} as the selection of stock s state in bit i. A value of 0 indicates the stock is not selected; a value of 1 indicates the stock is selected. The encoding length is the number of stocks.

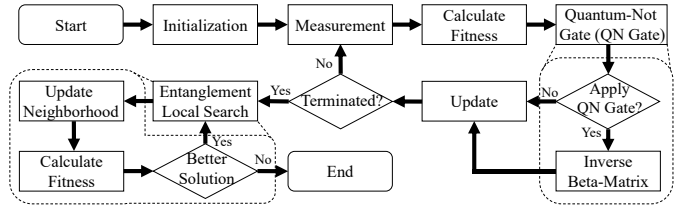

Fig. 1: Flow chart of EL-GNQTS.

Stock	Stock A	Stock B	Stock C	Stock D	Stock E
Beta-Matrix (b_{si})	0.5	0.5	0.5	0.5	0.5
Random Number	0.3	0.7	0.1	0.5	0.6
Solution (S_{si})	1	0	1	1	0

Fig. 2: Initialization of the beta-matrix and measurement of the status of a portfolio.

2) Initialization: EL-GNQTS is a probability-based algorithm that utilizes the concept of superposition, as shown in Equation (5). EL-GNQTS uses beta-matrix (b_{si}) to represent the probability of stock s in a bit i being selected in superposition. In the initialization phase, Fig. 2 shows that EL-GNQTS uses 0.5 as the initial probability, meaning that each stock has a probability of 50% of being selected into the portfolio.

$$|\psi\rangle = \alpha|0\rangle + \beta|1\rangle, \text{where } \alpha^2 + \beta^2 = 1 \tag{5}$$

3) Measurement: EL-GNQTS uses b_{si} and a random number to select stocks in the composite portfolio: if the probability in b_{si} is larger than a random number, EL-GNQTS will select the stock. Equation (6) and Fig. 2 show the generated process of solution (S) by comparison with the beta-matrix. The portfolio P is composed of selected stocks, where $P = \{stock(S_{si})|S_{si} = 1, si \in [1, n]\}$.

$$S_{si} = \begin{cases} 1, & \text{When } b_{si} \geq Random\ Number \\ 0, & \text{When } b_{si} < Random\ Number \end{cases} \tag{6}$$

4) Quantum-Not Gate Mechanism: The mechanism of the quantum-not gate ($X\ gate : \alpha\,|0\rangle + \beta\,|1\rangle \xrightarrow{X\ gate} \beta\,|0\rangle + \alpha\,|1\rangle$) used in EL-GNQTS is an important improvement of the algorithm. It helps the algorithm escape local optima toward the global best solution rather than a random position. It is used to invert the beta-matrix when the probability in the beta-matrix is higher than 0.5, but if it is not selected in the portfolio, the quantum-not gate will be applied to invert the probability in the beta-matrix to continue tracking the global best portfolio solution (Gb_{si}), and vice versa. Equation (7) and Fig. 3 show the quantum-not gate equation and operation.

$$If \begin{cases} Gb_{si} = 1\ \&\ Lw_{si} = 0\ \&\ b_{si} < 0.5 \\ Gb_{si} = 0\ \&\ Lw_{si} = 1\ \&\ b_{si} > 0.5 \end{cases},\ b_{si} = 1 - b_{si} \tag{7}$$

5) Update: The update is the most important phase in EL-GNQTS. We use the TR as the fitness metric, where higher values are better. EL-GNQTS uses Gb_{si} in all generations and the local worst solution (Lw_{si}) in the current generation, both of which correspond to stock s of bit i of the portfolio. If the global best solution and the local worst solution have different statuses, the beta-matrix is updated by θ, a parameter that can be defined based on the user's requirements. Equation (8) shows the update condition of the algorithm, and Fig. 3 shows details of the update operation.

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Stock	Stock A	Stock B	Stock C	Stock D	Stock E
Global Best (Gb_{si})	1	0	1	0	0
Local Worst (Lw_{si})	0	1	0	1	0
Beta-Matrix (b_{si})	0.6	0.3	0.4	0.8	0.5
Quantum-Not Gate					
Inversed Matrix	0.6	0.3	0.6	0.2	0.5
Update Operation (θ=0.1)					
Updated Matrix	0.7	0.2	0.7	0.1	0.5

Fig. 3: The illustration demonstrates the quantum-not gate and update mechanics, inverting and updating θ in the beta-matrix probability.

$$b_{si} = \begin{cases} b_{si} + \theta, & \text{When } Gb_{si} = 1 \ \& \ Lw_{si} = 0 \\ b_{si} - \theta, & \text{When } Gb_{si} = 0 \ \& \ Lw_{si} = 1 \\ b_{si}, & \text{When } Gb_{si} = Lw_{si} \end{cases} \tag{8}$$

6) Entanglement-enhanced Local Search Mechanism: Local search is a mechanism used in evolutionary computation to improve an algorithm's ability to search the neighborhood of a potential area. Entanglement-enhanced local search is proposed for portfolio optimization in this study. Through the detailed analysis of the solution relation, this study observes that the portfolio relationship can be seen as the entanglement relation between stocks. For example, two competing companies, A and B, may exhibit correlated growth or decline in a period. When there is a portfolio with B, the mechanism searches for stock A to replace stock B as a possible solution, such as $|\psi^+\rangle = \frac{1}{\sqrt{2}}(|01\rangle + |10\rangle)$. On the other hand, the stocks may be present or absent at the same time, such as $|\phi^+\rangle = \frac{1}{\sqrt{2}}(|00\rangle + |11\rangle)$. The ordinary neighbors are portfolio combinations with the least difference, i.e., the Hamming distance is 1, where portfolios include one additional stock or exclude one selected stock. The proposed mechanism introduces the novel neighborhood in portfolios based on the entanglement relation and then searches more precisely than randomly for the potential area. The entangled neighborhood includes a Hamming distance greater than one and the cardinality number $|P|$ (the number of selected stocks) with the smallest or unchanged difference. The potential neighborhood can produce according to Equation (9). The neighborhood around the global best portfolio S_{Gb} denotes as S_{nb}. This study considers three entangled neighborhood relations: unchanged (+1-1) and least (+2-1 and +1-2) cardinalities. The entanglement relation reduces the search degree of freedom and improves search effectiveness more than random. When a problem involves multiple dimensions of dependency, the proposed EL-GNQTS can enhance the solution to a certain extent. Thus, the entanglement-enhanced local search is a worthy developed technique.

$$\begin{matrix}\text{Entangled} \\ \text{Neighborhood}\end{matrix} \in \begin{cases} \sum_{si=1}^{n}(S_{si}^{Gb} \oplus S_{si}^{nb}) = 3 \text{ and } |(|P^{Gb}| - |P^{nb}|)| = 1 \\ \sum_{si=1}^{n}(S_{si}^{Gb} \oplus S_{si}^{nb}) = 2 \text{ and } |P^{Gb}| = |P^{nb}| \end{cases} \tag{9}$$

C. Sliding Window

There are thousands of historical stock price data. Adopting years of data as a training period to invest for a day or a week is unreasonable. Therefore, it is important to find a suitable training-to-testing period. We adopt common financial report periods, a year (Y), a half year (H), a quarter (Q), and a month (M), respectively, to divide investment length into 13 training-to-testing types: Y2Y, Y2H, Y2Q, Y2M, H2H, H2Q, H2M, Q2Q, Q2M, M2M, H*, Q*, and M*; e.g., Y2Q refers to training for a year and then testing in the next quarter. The H*, Q*, and M* train from the same prior year period for training to consider the business cycle. These sliding windows help us maintain the data's freshness and remove outdated data.

TABLE I: The experimental parameters of EL-GNQTS

Initial Fund	Runs	Populations	Iterations	θ
10 Millions Local Currency	50	10	10,000 for Most Experiments 20,000 for Global G7 Markets	0.0002

IV. Experimental Results

This section introduces the experimental environment first and then compares the performance of the proposed intelligent system with that of the Sharpe ratio, the indexes, and the benchmark uniform buy & hold (UBAH) [25] strategy to prove the universality of the trend ratio strategy among G7 markets. After demonstrating our strategy's effectiveness and generality, an in-depth investigation and analysis of the different countries is conducted. Finally, a global asset management system integrating G7 markets is considered to facilitate more robust investment through trend ratio evaluation.

A. Experimental Environment

The investment period includes ten years, from Jan. 2013 to Dec. 2022, with 831 rebalancing training-to-testing periods. The investment stock markets in the experiments are those of the G7 members. The G7 is an intergovernmental political forum focusing on economic strategy and accounts for half of the global net wealth, making it representative.

The experiments are conducted on the iconic markets of the G7, including the Dow Jones Industrial Average (DJIA) in the United States (U.S.), the S&P Toronto Stock Exchange Composite Index (TSX) in Canada, the Financial Times Stock Exchange 100 index (FTSE) in the United Kingdom (U.K.), Cotation Assistée en Continu 40 (CAC) in France, Deutscher Aktienindex (DAX) in Germany, FTSE Milano Indice di Borsa Index (MIB) in Italy, and Nikkei 225 (NIKKEI) in Japan. We select the top 30 large market-cap companies based on complete data from constituents over ten years. The market-cap ranking data are referenced from TradingView. The stock price data are from Yahoo Finance, Investing.com, and Wharton Research Data Services (WRDS). The selected stocks in a portfolio are weighted equally. Table I shows the experimental parameters of the proposed intelligent strategy, where θ is the most important parameter in EL-GNQTS. To achieve robustness, the fine-tuned procedure displays the performance of the M2M sliding window using the average trend ratio of 50 runs, as shown in Fig. 4. This study fine-tunes θ in different markets. It obtains a consistent result: 0.0002 has the best result. Other parameter settings are selected from [8]. For the following comparison of strategies, each period selects the best solution from 50 runs, and the provided result in each country is the mean of the 831 periods.

B. Self-Analysis Between Portfolios and Single Stocks

First, this study compares the performance of risk and trend ratio between the portfolios and single stocks. The single stock has the most stable uptrend in each period, one of the 30 companies with the highest trend ratio. The portfolio selected by EL-GNQTS is one with a stable uptrend combination in each period.

Fig. 5(a) shows the portfolio performance and the single stocks among G7 markets regarding risk. In all markets, the portfolio risk is lower than that of a single stock. The result proves that the portfolio diversifies the risk more effectively

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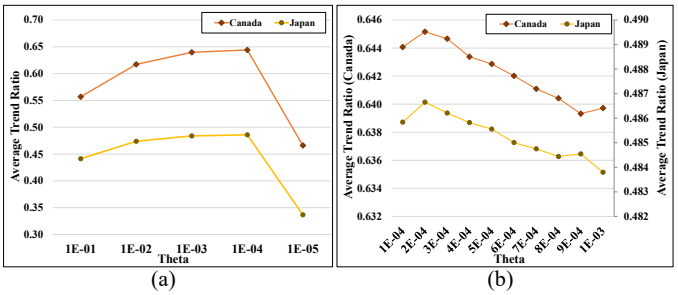
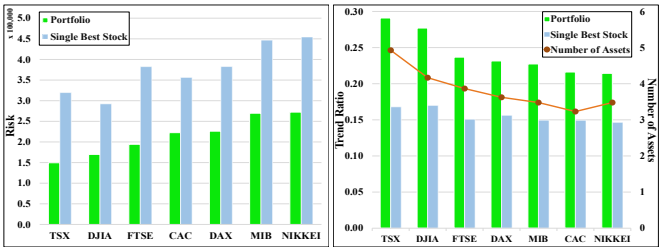


(a) (b)

Fig. 4: The θ fine-tuned process of the various markets is consistent, where 1E-04 to 1E-03 is a better range, and 0.0002 is the best θ.



(a) Risk comparison (b) Trend ratio comparison

Fig. 5: Comparisons are between the portfolio and the single best stock performance. The ranking is according to the portfolio in ascending order. (a) The portfolio risk is lower than that of a single stock. The TSX and DJIA achieve the prior two ranks regarding the risk in the portfolio and single stock. (b) The portfolio trend ratio is higher than that of a single stock. The number of assets and the trend ratio are strongly related.

than a single stock in all G7 markets; as the ancient saying goes, "Do not put all your eggs in one basket." Regarding the portfolio and single stock risk, the TSX and DJIA rank in the top 2 places. We observe more detail in these two markets. In Canada, the risk of a single stock is higher than that in the U.S., but the risk of a portfolio is inverted. We believe that the industries in Canada have a higher degree of complementarity, helping reduce portfolio risk. Investing in a single stock is accompanied by high risk, while a portfolio can diversify the risk effectively. In addition, the U.S. has the lowest risk of a single stock because the U.S. stock market is a representative financial indicator and has the largest exchanged market value in the world. Therefore, it appeals to many investors worldwide; thus, the liquidity is high to trade the security in a short time, which leads to a decrease in the risk of a single stock.

The trend ratio is a metric that can be used to find a perfect trade-off between return and risk, i.e., the daily expected return per unit of daily risk. Fig. 5(b) displays the trend ratio performance between the portfolio and single stock among the G7 markets. The results indicate that the trend ratio of a portfolio is higher than that of a single stock in all G7 markets. The TSX portfolio has the highest trend ratio and lowest risk. The ranking of trend ratio is highly correlated with the risk ranking; thus, risk dispersion is a critical issue in balancing return and risk performance. In most cases, markets with a higher trend ratio are composed of a greater number of stocks: the stock number is proportional to the trend ratio. A higher number of assets decreases portfolio risk and promotes the performance of the trend ratio.

C. Comparison with the Sharpe Ratio, Index, and Benchmark Strategy

This section compares the proposed method with the Sharpe ratio, the benchmark UBAH strategy, and indexes in terms of the risk, trend ratio, maximum drawdown (MDD) [26], and profit factor (PF). The results reported for each country are the average of 831 rebalancing periods.

MDD and PF are two indicators commonly used to evaluate investment strategy performance. MDD assesses the maximum loss risk from the portfolio peak to trough, $MDD = max_{\tau\in(0,td)}(\frac{max_{t\in(0,\tau)}FS_t - FS_\tau}{max_{t\in(0,\tau)}FS_t})$. A lower MDD represents lower volatility, meaning the portfolio has a more stable trend and less loss from the peak. PF is a return metric calculated as the gross profit (GP) divided by the gross loss (GL), which means a positive return per unit of negative return. The GP and GL in this study are calculated by daily FS. The experimental results in Fig. 6(c) show that our strategy achieves a substantially lower MDD than the other strategies in each market.

The UBAH equally weights each stock in each investment target. The index indicates the performance of the corresponding market index. As shown in Fig. 6(a), the risk of a portfolio guided by our strategy has superior performance compared to other strategies. Fig. 6(b) shows that our strategy outperforms each market's Sharpe ratio, UBAH, and representative index. This demonstrates that our proposed strategy performs better in identifying portfolios with a high return per unit of risk. The results in Fig. 6(d) show that the Sharpe ratio performs better than UBAH and index strategies with more than 1 PF value. Moreover, our strategy performs superior to the Sharpe ratio and achieves greater than 2 in all markets. In the DJIA and NIKKEI markets, our strategy achieves a PF value greater than 3, which indicates that our proposed system for assessing promising portfolios outperforms the other three strategies.

The DJIA has a low MDD and a high PF, indicating a low risk and high return. The DJIA attains the second-best trend ratio performance with a balanced return and risk. The analysis indicates that the trend ratio evaluation is consistent with the other representative indicators. When the performance in terms of MDD (risk) and PF (return) improves, the return per unit of risk also improves, which is the meaning of the trend ratio. The TSX performs best in terms of risk, MDD, and trend ratio, which means the risk accounts for a large proportion of the effect when elevating the portfolio performance.

In the strategy comparison, we observe that although the portfolio can effectively spread risk and elevate the investment efficiency over those of a single stock, taking all stocks into account is not the best choice, as indicated by the UBAH and the index performance. An effective strategy and an intelligent system that aid in creating a valuable portfolio are essential for spreading risk and increasing return per unit of risk.

Fig. 7 illustrates the FS trend among the different strategies. Our green line strategy indicates a stable uptrend, and the average line limits the Sharpe ratio. The UBAH and the representative index have a significant risk and low profit compared to our strategy. Our portfolio has a stronger uptrend and lower volatility than others. Thus, our constructed portfolio has a high trend ratio, low risk, low MDD, and high PF.

To verify the significant improvement of the proposed strategy compared to the other three strategies, we conduct

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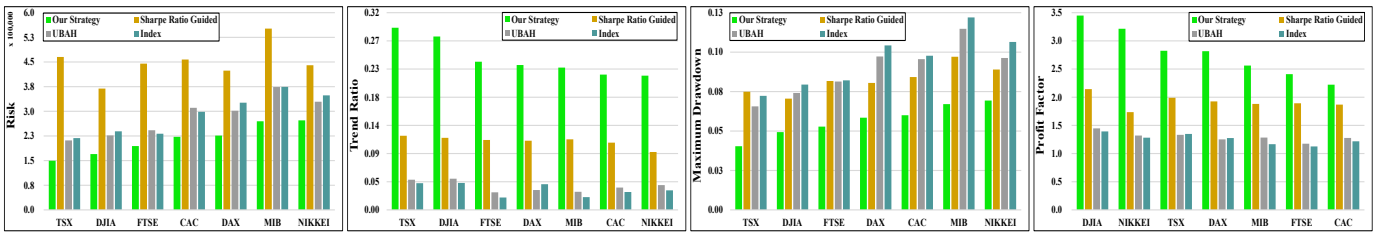

(a) Risk comparison (b) Trend ratio comparison (c) Maximum drawdown comparison (d) Profit factor comparison

Fig. 6: Comparison of portfolios guided by our strategy, SR, UBAH, and representative index among the G7 markets. Our strategy (green) outperforms other strategies regarding risk, trend ratio, maximum drawdown, and profit factor.

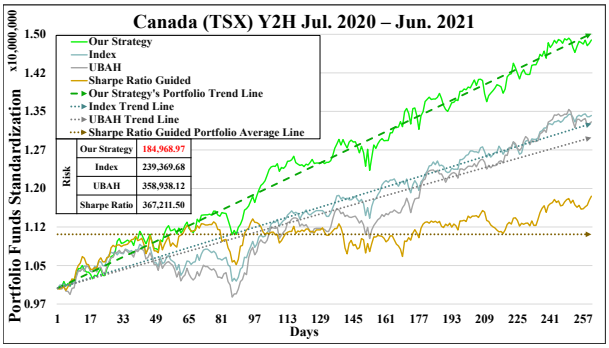

Fig. 7: Comparison of portfolios guided by our strategy, SR, UBAH, and representative index in FS. Our portfolio has a stable uptrend and lower volatility.

TABLE II: One-tailed t-tests are conducted to compare our portfolio (O) with the Sharpe ratio (S), UBAH (U), and indexes (I) in terms of the risk and trend ratios at the 5% significance level.

Markets	TSX	DJIA	FTSE	DAX	MIB	CAC	NIKKEI	
Null Hypothesis (H_0)	p-value	p-value	p-value	p-value	p-value	p-value	p-value	Result
$\mu^O_{TR} \leq \mu^S_{TR}$	4.22E-48	1.39E-37	2.11E-31	1.85E-31	1.33E-25	6.55E-27	4.12E-37	All Reject H_0
$\mu^O_{Risk} \geq \mu^S_{Risk}$	4.29E-51	2.61E-73	1.85E-62	3.07E-52	2.53E-54	2.38E-43	7.47E-33	All Reject H_0
$\mu^O_{TR} \leq \mu^I_{TR}$	5.29E-95	1.09E-73	9.01E-89	2.11E-72	9.10E-78	3.42E-77	4.01E-74	All Reject H_0
$\mu^O_{Risk} \geq \mu^I_{Risk}$	2.16E-17	9.93E-18	9.87E-07	3.49E-25	1.03E-18	2.13E-14	2.05E-12	All Reject H_0
$\mu^O_{TR} \leq \mu^U_{TR}$	2.32E-91	1.40E-69	2.95E-82	3.88E-81	1.45E-71	9.95E-72	6.72E-69	All Reject H_0
$\mu^O_{Risk} \geq \mu^U_{Risk}$	3.03E-15	6.03E-14	1.82E-09	7.93E-14	3.24E-18	1.23E-17	6.95E-08	All Reject H_0

hypothesis tests comparing our portfolio (O) with the Sharpe ratio (S), UBAH (U), and indexes (I) in terms of the risk and the trend ratio at the 5% significance level. Table II presents the results of one-tailed t-tests for each G7 market. μ represents the average value in each tested sequence. The null hypotheses of the risk and the trend ratio for each test are $\mu^{\mathrm{O}}_{\mathrm{Risk}} \geq \mu^{\mathrm{S/U/I}}_{\mathrm{Risk}}$ and $\mu^{\mathrm{O}}_{\mathrm{TR}} \leq \mu^{\mathrm{S/U/I}}_{\mathrm{TR}}$. With a single null hypothesis, all hypothesis testing is conducted independently. All the p-values are less than the significance level of 0.05. The null hypotheses are all rejected. Regarding the trend ratio performance, as the gap between our strategy and the index is the greatest in Fig. 6(b), the p-value of null hypothesis $\mu^O_{\mathrm{TR}} \leq \mu^I_{\mathrm{TR}}$ is the lowest.

D. General and Special Case Analysis Among Markets

We conduct a detailed self-analysis in this section. The discussion focuses on the similarities and differences in each market under the trend ratio evaluation regarding the best portfolio trending and the constituents. Due to the limited space, there are listed major and general discussion cases.

Regarding the best portfolio composition, over 82% of periods contain the first-ranked trend ratio stock, as shown in Table III. Fig. 8 depicts the FS fluctuation between the portfolio and its constituents, including the first-ranked stock. The proposed model by trend ratio is effective for each market. The portfolio contains many constituents that can effectively diversify the risk, making FS close to the trend line. According to the figure, the portfolio with the best trend ratio selected by EL-GNQTS has a more stable uptrend with low risk and high return compared to the best single stock. Thus, a portfolio constructed based on the trend ratio achieves superior results.

TABLE III: Selection rate of the first-ranked stocks of G7 members.

Markets	DJIA	TSX	FTSE	MIB	NIKKEI	DAX	CAC
Rate	82.77%	86.63%	86.99%	88.07%	89.16%	89.40%	90.48%

TABLE IV: Verification of the case of the portfolio shown in Fig. 11, which contains a stock with a negative return (NR) and excludes the first-ranked stock.

Neighborhood Subset	Components	Trend Ratio	Risk
Our Portfolio	6702.T(2), 6367.T(3), 8058.T(4), 8035.T(7), 6954.T(13), 4519.T(30)	**0.1413416**	**161,677.98**
Add 1st Stock	8031.T(1), 6702.T(2), 6367.T(3), 8058.T(4), 8035.T(7), 6954.T(13), 4519.T(30)	0.1366257	177,632.60
Remove NR Stock	6702.T(2), 6367.T(3), 8058.T(4), 8035.T(7), 6954.T(13)	0.1305992	231,561.11
Add 1st Stock Remove NR Stock	8031.T(1), 6702.T(2), 6367.T(3), 8058.T(4), 8035.T(7), 6954.T(13)	0.1230411	249,286.60

A more detailed observation of the constituents indicates three interesting homogeneous phenomena in most G7 markets. First, the first-ranked stock is sometimes excluded, but the portfolio still has a stable uptrend with low risk and high return, as shown in Fig. 9. Second, a stock with a negative return is sometimes selected to strike a balance between risk and return in the best portfolio, as shown in Fig. 10. The trend ratio attaches great importance to the corresponding high risk while seeking high returns. Third, the best portfolio may contain a stock with a negative return and exclude the first-ranked stock, as shown in Fig. 11. A stock with a negative return further spreads the risk. The constituents in a portfolio are commonly interrelated; if one of the components has a high return, it will be matched with another component to disperse the risk. To verify these interesting phenomena and the ability of EL-GNQTS, Table IV displays a comparison of the best-searched trend ratio and other portfolios from Fig. 11. The verification involves a subset of the entanglement-enhanced local search near the local optimum, including the neighbors with the first ranking, without a negative return, and containing both states.

E. Global Asset Financial Management

According to the evidence related to a series of effects of the global COVID-19 pandemic, global markets, and economies significantly impact one another. Investment targets are no longer bound by borders or restricted by local markets due to the emergence of online brokers and trading platforms like eToro, TD Ameritrade, E*TRADE, Firstrade, and Robinhood. Therefore, more robust investment can be achieved by trading

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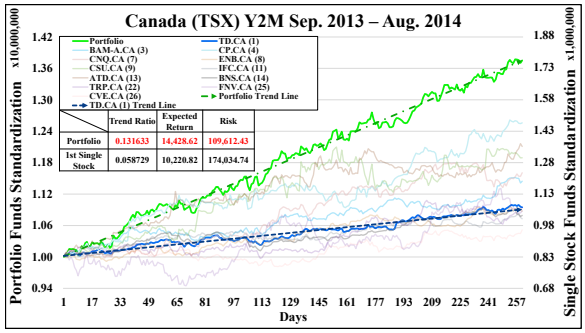
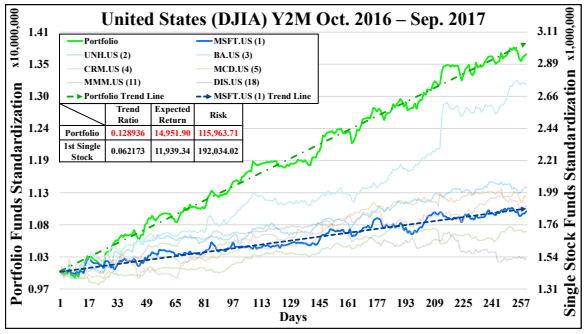
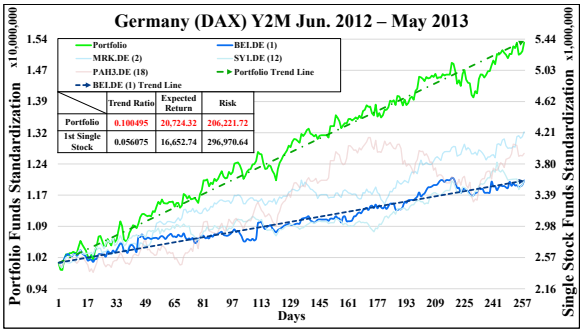


(a) TSX



(b) DJIA



(c) DAX

Fig. 8: The best portfolio (green) comprises multiple constituents, including the first-ranked stock. In these periods, the portfolio with a stable uptrend has a better trend ratio, expected return, and risk performance than the first-ranked stock.

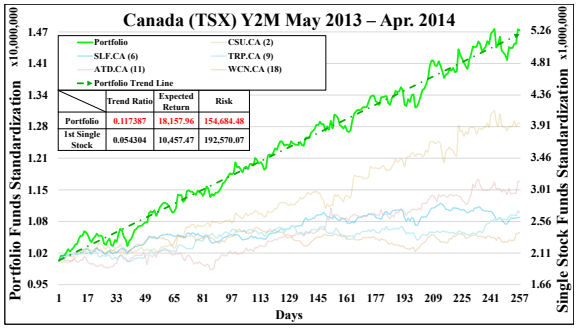


Fig. 9: The best portfolio (green) excludes the first-ranked stock in TSX. The portfolio outperforms the first-ranked stock in terms of the trend ratio, return, and risk performance.

in global markets, especially the stock markets of the G7, which account for almost 60% of the market value of the global stock exchange. Investing in various industries across markets benefits investors by creating a complementary portfolio and minimizing systematic risk. This study proposes an innovative global asset management model based on the novel trend ratio and the effective EC, EL-GNQTS. This study is the first attempt to consider global markets in portfolio optimization models based on the trend ratio.

The rebalancing and training in a month are the most

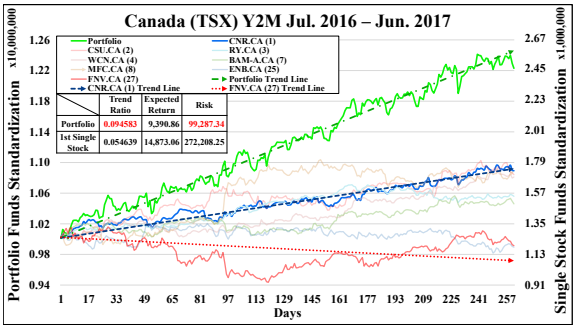


Fig. 10: The best portfolio (green) includes both the first-ranked stock and a stock with a negative return in TSX. Compared to those of the best single stock, the risk in the portfolio decreased by almost one-third and the trend ratio increased by a factor of almost two.

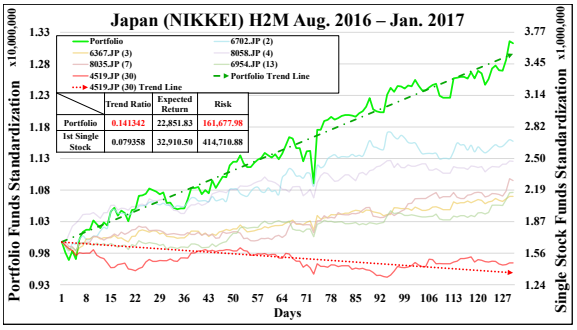


Fig. 11: The best portfolio (green) contains a stock with a negative return and excludes the first-ranked stock. The stock with a negative return is the last-ranking stock, which is to counteract the risk. The trends of the stocks ranked 2nd and 30th show an inverse trend.

challenging, and thus the comparison among the EL-GNQTS and other state-of-the-art optimization techniques GNQTS [8], GQTS [27], QTS [5], and the GA [11] focuses on the M2M and M*. The parameters of the GNQTS, GQTS, and QTS techniques are the same as in Table I. GA employs the same process and parameters as [11] fine-tuned: 100 chromosomes, 0.6% mutation rate of a single bit, and a 20% preserved rate. The first comparison integrates the top 2 ranked markets, TSX and DJIA, into an investment target. The solution space is 2^{60}. In Table V, the EL-GNQTS outperforms other techniques significantly, as indicated by the ranking EL-GNQTS>GNQTS>GQTS>QTS>GA. This result of the EL-GNQTS takes an average of 456 additional evaluations to produce better solutions.

EL-GNQTS and GNQTS are compared in greater detail in an expanded solution space by incorporating 90 stocks from three markets (TSX, DJIA, and Nikkei). The proposed EL-GNQTS reveals exceptional portfolio performance compared to the GNQTS in Table VI, outperforming the GNQTS in all sliding window cases and requiring an average of only 1,522 additional evaluations. To determine the significance of Table VI, the hypothesis test is conducted at a significance level of 5%, and the p-value in Table VII, is less than 0.05, indicating that EL-GNQTS has a significant improvement over GNQTS in this comparison. Moreover, the convergent analysis employs the same 30,000 evaluations for each algorithm. The convergence graph is the average value of 50 runs to demonstrate the algorithm's robustness. Fig. 12(a) demonstrates that the GNQTS and EL-GNQTS outperform other algorithms, where the GA converges quickly. In Fig. 12(b), a detailed analysis of 10,000 to 30,000 evaluations reveals that EL-GNQTS has robust growth potential, whereas GNQTS has converged

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TABLE V: EL-GNQTS compared with other state-of-the-art algorithms, GNQTS, GQTS, QTS, and GA.

Algorithm	EL-GNQTS	GNQTS	GQTS	QTS	GA
M*	**1.024848**	1.024353	0.953591	0.722358	0.695244
M2M	**1.005041**	1.004546	0.93214	0.706025	0.671347

TABLE VI: Trend ratio comparison between EL-GNQTS and GNQTS for 13 sliding windows.

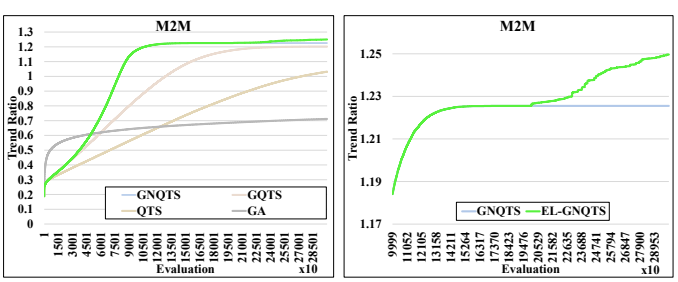
Algorithm	M*	M2M	Q*	Q2Q	Q2M	H*	H2H
EL-GNQTS	**1.355364**	**1.332698**	**0.393991**	**0.379458**	**0.370615**	**0.199167**	**0.198045**
GNQTS	1.340680	1.318418	0.393415	0.378882	0.369675	0.198580	0.197458
Algorithm	H2Q	H2M	Y2H	Y2Q	Y2Y	Y2M	Overall
EL-GNQTS	**0.194257**	**0.186180**	**0.101597**	**0.100246**	**0.100167**	**0.098325**	**0.547507**
GNQTS	0.193775	0.185805	0.101031	0.099843	0.099773	0.097949	0.542935

In this experiment, all the constituents of the G7 are integrated into a global portfolio solution space, which can be used to reduce the systematic risk considerably. There are 210 stocks in the global asset selection. The global market can help investors invest more robustly. Currently, international trade is common and frequent. Different markets influence one another. A high-profit stock has a greater chance of discovering a complementary stock to achieve low risk and high trend ratio when multiple markets are integrated. The risk performance of the global market is lower than any single G7 market, as shown in Fig. 13, and the trend ratio for global portfolio construction outperforms a single market. Thus, EL-GNQTS is an effective and efficient EC for constructing portfolios by exploring 10E-59 times on average compared to the entire solution space.

The global portfolio is constructed from a diverse selection of global stocks, and portfolios consisting of worldwide stocks outperform the best single stocks of G7 members. The global portfolios contain more stocks and achieve more stable uptrends, as shown in Fig. 14, where the portfolio trend ratio is triple that of single stocks and the risk is less than one-half of that of the best single stock. The experimental results illustrate that EL-GNQTS can find efficient portfolios with stable uptrends and low volatility, which satisfies investors' expectations perfectly. In a more detailed analysis in Fig. 15, we observe that Canadian and American stocks account for most of the global portfolio. The third most frequently chosen stock is Japanese stock. Therefore, the global portfolio must consider stock complementary relationships to compose a stable trend. Each market contributes to a different degree to improve the trend ratio performance.

TABLE VII: The hypothesis test result of the improvement.

Null Hypothesis (H_0)	p-value	Result
$\mu_{EL} = \mu_{GN}$	7.75E-11	Reject H_0
$\mu_{EL} \leq \mu_{GN}$	3.88E-11	Reject H_0



(a) (b)

Fig. 12: Based on the same evaluations in a larger solution space that includes three markets, the results are average in 50 runs. EL-GNQTS has the best performance and continually improves, whereas other algorithms converge gradually.

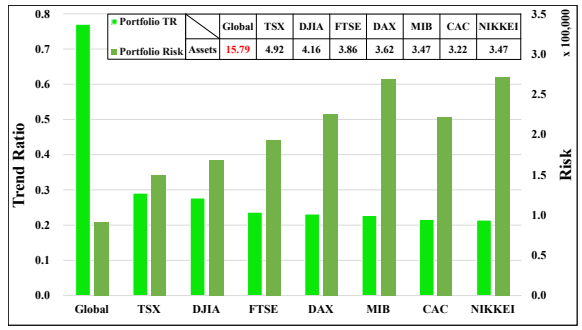


Fig. 13: Comprehensive comparison of global intermarket and single-country investing. Global market investing overwhelmingly outperforms single-country investing regarding both the trend ratio and risk.

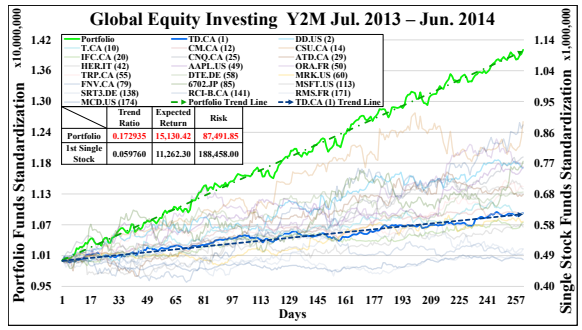


Fig. 14: The high-performing global market portfolio has significantly less volatility. The portfolio has a better trend ratio, expected return, and risk than the best stock.

F. Discussion

With the development of effective global financial markets and based on our results, we believe that the global market can be used to create uptrend portfolios with complementary trends; thus, the market relationship is an important topic of discussion. This study conducts a preliminary test to investigate the relationships among markets and their efficiency in improving the performance of the trend ratio. Based on the respective trend ratio performance ranking, the experiment gradually expanded to the global market. The experiment aims to verify whether a solution space with more stocks is beneficial to achieving a higher trend ratio. Fig. 16 indicates the targets initially include one market. Then, one country at a time is added to the solution space to improve the performance of the trend ratio. Finally, although the trend ratio increases when considering seven markets, marginal utility arises. The greatest increase occurs from the first to the second market, indicating that a portfolio from multiple markets can significantly increase investment stability. This study shows that the proposed global financial asset management system based on trend ratios can successfully increase the effectiveness of investments. As the market size increases, the solution space increases by 2^{30} each time. The experiment exhibits

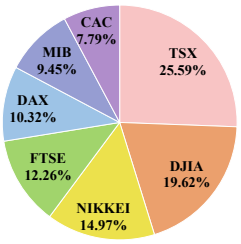


Fig. 15: Illustration of the proportion of G7 markets in the global portfolio.

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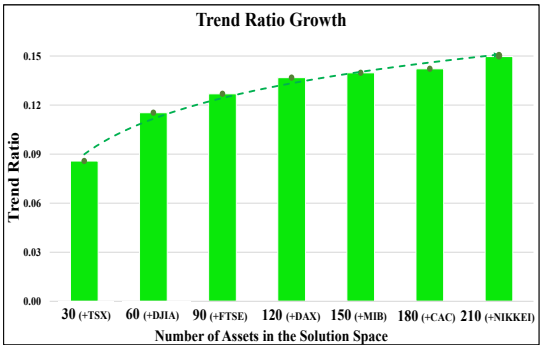

Fig. 16: Including a greater number of global markets can yield a higher trend ratio as more possibly complementary industries are added, but the extent of the improvement gradually declines.

promising results regarding considering market relationships to compose a better portfolio with an effective EL-GNQTS EC. The relationship among different markets is worthy of discussion and can be improved by the EC techniques.

V. Conclusion

The proposed EL-GNQTS defines a novel entangled neighborhood relation for portfolio optimization to apply the precise and effective local search rather than random searched, which has not been previously explored. EL-GNQTS considers the entangled relationship between stocks to effectively reduce the search degree of freedom, thereby enhancing search capability. Using the EL-GNQTS, the trend ratio can be used to robustly identify a stable uptrend portfolio in major world markets. In addition, this study is the first attempt to expand the generality of using the trend ratio in major industrialized global G7 markets by conducting an in-depth investigation of portfolio construction. The novel evolutionary computation technique EL-GNQTS applies to a real-world financial application, achieving excellent performance and thoroughly analyzing the results. The proposed intelligent system outperforms the Sharpe ratio, UBAH, and index strategies regarding risk, trend ratio, MDD, and PF. Statistical hypothesis testing illustrates that our proposed method significantly improves compared to these targets regarding the trend ratio and risk performance. Furthermore, this study incorporates global markets as an investment objective. As the solution space expands, EL-GNQTS is shown to be more effective than others.

The results indicate that a global portfolio has an outstanding performance in achieving higher returns per unit of risk. Moreover, the trend ratio plays a general and effective role in the proposed innovative global portfolio optimization model. This study discovers many interesting phenomena in promoting investment performance while considering different combinations of markets. When different markets interact, they each contribute to improving investment efficacy to a different extent. Global asset management is a practical consideration in the current global economy. The proposed intelligent system provides an avenue for globally diversified investment based on the trend ratio indicator. The trend ratio performance can be generalized and explored with other optimizers in future work, such as other quantum optimization techniques. In the near future, quantum computing will aid in high-efficiency searching for optimal solutions in our proposed global investment model with the innovative trend ratio indicator, addressing more stock relationships effectively.

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